

3.1.1 Applied anatomy and physiology

Students should develop knowledge and understanding of the key body systems and how they impact on health, fitness and performance in physical activity and sport.

3.1.1.1 The structure and functions of the musculoskeletal system

Content	Additional information
Bones	Identification of the bones at the following locations: • head/neck – cranium and vertebrae • shoulder – scapula and humerus • chest – ribs and sternum • elbow – humerus, radius and ulna • hip – pelvis and femur • knee – femur and tibia (students should also know that the patella sits in front of the knee joint) • ankle – tibia, fibula and talus.
Structure of the skeleton	How the skeletal system provides a framework for movement (in conjunction with the muscular system): • the skeletal system allows movement at a joint • the shape and type of the bones determine the amount of movement (short bones enable finer controlled movements/ long bones enable gross movement) • flat bones for protection of vital organs • the different joint types allow different types of movement • the skeleton provides a point of attachment for muscles – when muscles contract they pull the bone.

Content	Additional information
Functions of the skeleton	• support • protection of vital organs by flat bones • movement • structural shape and points for attachment • mineral storage • blood cell production. Functions should be applied to performance in physical activity.
Muscles of the body	Identification of the following muscles within the body: • latissimus dorsi • deltoid • rotator cuffs • pectorals • biceps • triceps • abdominals • hip flexors • gluteals • hamstring group (not individual names) • quadriceps group (not individual names) • gastrocnemius • tibialis anterior. Students should be taught the role of tendons (attaching muscle to bones).
Structure of a synovial joint	Identification of the following structures of a synovial joint and how they help to prevent injury: • synovial membrane • synovial fluid • joint capsule • bursae • cartilage • ligaments.
Types of freely movable joints that allow different movements	Identification of the types of joints with reference to the following: • elbow, knee and ankle – hinge joint • hip and shoulder – ball and socket.

Content	Additional information
How joints differ in design to allow certain types of movement at a joint	Understand that the following types of movement are linked to the appropriate joint type, which enables that movement to take place: • flexion/extension at the shoulder, elbow, hip and knee • abduction/adduction at the shoulder • rotation of the shoulder • circumduction of the shoulder • plantar flexion/dorsiflexion at the ankle. Application to specific sporting actions is in movement analysis (page 16).
How the major muscles and muscle groups of the body work antagonistically on the major joints of the skeleton to affect movement in physical activity at the major movable joints	With reference to the shoulder, elbow, hip, knee and ankle joints: • major muscle groups operating at these joints (see above) • the action of prime movers (agonists)/ antagonists • bones located at the joint (see above) • how these muscle groups work isometrically and isotonicity (concentric/ eccentric). The difference between concentric and eccentric (isotonic) contractions.

3.1.1.2 The structure and functions of the cardio-respiratory system

Content	Additional information
The pathway of air	Identification of the pathway of air (limited to): • mouth/nose • trachea • bronchi • bronchioles • alveoli.

Content	Additional information
Gaseous exchange	Gas exchange at the alveoli – features that assist in gaseous exchange: • large surface area of alveoli • moist thin walls (one cell thick) • short distance for diffusion (short diffusion pathway) • lots of capillaries • large blood supply • movement of gas from high concentration to low concentration. Oxygen combines with haemoglobin in the red blood cells to form oxyhaemoglobin. Students should also know that haemoglobin can carry carbon dioxide.
Blood vessels	Structure of arteries, capillaries and veins: • size/diameter • wall thickness • valves in veins. How the structure of each blood vessel relates to the function: • carrying oxygenated/deoxygenated blood to/from the heart • gas exchange • blood pressure • redistribution of blood during exercise (vasoconstriction and vasodilation). Students should be taught the names of the arteries and the veins associated with blood entering and leaving the heart.
Structure of the heart	Structure of the heart: • atria (left and right atria) • ventricles (left and right ventricles).

Content	Additional information
The cardiac cycle and the pathway of the blood	The order of the cardiac cycle, including diastole (filling) and systole (ejection) of the chambers. This starts from a specified chamber of the heart, eg the cardiac cycle starting at the right ventricle. Pathway of the blood: • deoxygenated blood into right atrium • then into the right ventricle • the pulmonary artery then transports deoxygenated blood to the lungs • gas exchange occurs (blood is oxygenated) • pulmonary vein transports oxygenated blood back to the left atrium • then into the left ventricle • before oxygenated blood is ejected and transported to the body via the aorta. Valve names are not required but students should be taught that valves open due to pressure and close to prevent backflow.
Cardiac output, stroke volume and heart rate	Cardiac output, stroke volume and heart rate, and the relationship between them. Cardiac output (Q) = stroke volume x heart rate. Students should be taught how to interpret heart rate graphs, including an anticipatory rise, and changes in intensity.

Content	Additional information
Mechanics of breathing – the interaction of the intercostal muscles, ribs and diaphragm in breathing	Inhaling (at rest) with reference to the roles of the: • intercostals • rib cage • diaphragm. Exhaling (at rest) with reference to the roles of the: • intercostals • rib cage • diaphragm. Lungs can expand more during exercise (inspiration) due to the use of pectorals and sternocleidomastoid. During exercise (expiration), the rib cage is pulled down quicker to force air out quicker due to use of the abdominal muscles. Changes in air pressure cause the inhalation and exhalation.
Interpretation of a spirometer trace	Identification of the following volumes on a spirometer trace and an understanding of how these may change from rest to exercise: • tidal volume • expiratory reserve volume • inspiratory reserve volume • residual volume. Interpretation and explanation of a spirometer trace (and continue a trace on paper) to reflect the difference in a trace between rest and the onset of exercise.

3.1.1.3 Anaerobic and aerobic exercise

Content	Additional information
Understanding the terms aerobic exercise (in the presence of oxygen) and anaerobic exercise (in the absence of enough oxygen)	Definition of the terms: • aerobic exercise • anaerobic exercise. Summary of aerobic exercise (glucose + oxygen → energy + carbon dioxide + water). Summary of anaerobic exercise (glucose → energy + lactic acid).

Content	Additional information
The use of aerobic and anaerobic exercise in practical examples of differing intensities	Link practical examples of sporting situations to aerobic or anaerobic exercise. Identification of the duration and/or intensity of a physical activity in order to identify and justify why it would be aerobic or anaerobic, eg marathon (aerobic), sprint (anaerobic).
Excess post-exercise oxygen consumption (EPOC)/oxygen debt as the result of muscles respiring anaerobically during vigorous exercise and producing lactic acid	Definition of the term EPOC (oxygen debt). An understanding that EPOC (oxygen debt) is caused by anaerobic exercise (producing lactic acid) and requires the performer to maintain increased breathing rate after exercise to repay the debt.
The recovery process from vigorous exercise	The following methods to recover from exercise, including the reasons for their use: • cool down – maintain elevated breathing rate/heart rate (blood flow), stretching, removal of lactic acid • manipulation of diet – rehydration, carbohydrates for energy • ice baths/massage – prevention of delayed onset muscle soreness (DOMS). Students should be taught to evaluate the use of these methods, justifying their relevance to different sporting activities.

3.1.1.4 The short and long term effects of exercise

Content	Additional information
Immediate effects of exercise (during exercise)	• hot/sweaty/red skin • increase in depth and frequency of breathing • increased heart rate.
Short-term effects of exercise (up to 36 hours after exercise)	• tiredness/fatigue • light headedness • nausea • aching/delayed onset muscle soreness (DOMS)/cramp.

Content	Additional information
Long-term effects of exercise (months and years of exercising)	• body shape may change • improvements in specific components of fitness • build muscle strength • improve muscular endurance • improve speed • improve suppleness • build cardio vascular endurance • improve stamina • increase in the size of the heart (hypertrophy) • lower resting heart rate (bradycardia). Students should be taught the components of fitness to understand the long term effects of exercise.